

```
1  import pandas as pd
2
3  stud =pd.read_excel('stud1.xlsx')
4  dep= pd.read_excel('dept1.xlsx')
5
6  stud1 = pd.DataFrame(stud)
7  stud1
8
9  dep1 = pd.DataFrame(dep)
10 dep1
11
12
13
```

PROBLEMS

OUTPUT

TERMINAL

...

 Python + v 

PS C:\Users\prem2\Git files\python> C:/Users/prem2/AppData/Programs/Python/Python314/python.exe

```
>>> stud1
   stud_id  name  dept_id  marks
0         1  john      10     85
1         2  sara      20     90
2         3  mike      10     70
3         4  anna      30     95
4         5   tom      40     60
>>> dep1 = pd.DataFrame(dep)
>>> dep1
   dept_id dept_name
0        10  science
1        20  commerce
2        30     arts
>>> 
```

```
1  import pandas as pd
2
3  stud =pd.read_excel('stud1.xlsx')
4  dep= pd.read_excel('dept1.xlsx')
5
6  stud1 = pd.DataFrame(stud)
7  stud1
8
9  dep1 = pd.DataFrame(dep)
10 dep1
11
12
13 combine = pd.merge(stud,dep,how='inner',on='dept_id')
14 combine
15
```

PROBLEMS OUTPUT TERMINAL ...

Python + - [] [X] ... []

```
>>> combine = pd.merge(stud,dep,how='inner',on='dept_id')
>>> combine
   stud_id  name  dept_id  marks  dept_name
0         1  john        10     85   science
1         2  sara        20     90  commerce
2         3  mike        10     70   science
3         4  anna        30     95     arts
>>> 
```

● Click to add a breakpoint

```
11
12
13 combine = pd.merge(stud,dep,how='inner',on='dept_id')
14 combine
15 💡
16 stud.rename(columns={'name':'student_name'})
17
18 dep.rename(columns={'dept_name':'department'})
19
20 stud[stud['marks']>80]
21
22
```

PROBLEMS OUTPUT TERMINAL ... Python + - [] [] [] [] []

```
>>> stud.rename(columns={'name':'student_name'})
  stud_id student_name dept_id marks
0        1         john      10    85
1        2          sara      20    90
2        3          mike      10    70
3        4          anna      30    95
4        5           tom      40    60
>>> dep.rename(columns={'dept_name':'department'})
  dept_id department
0        10    science
1        20   commerce
2        30        arts
>>> [ ]
```



```

11
12
13 combine = pd.merge(stud,dep,how='inner',on='dept_id')
14 combine
15
16 stud.rename(columns={'name':'student_name'})
17
18 dep.rename(columns={'dept_name':'department'})
19
20 stud[stud['marks']>80]
21
22 💡
23 avg = stud.groupby('dept_id')['marks'].agg(['mean'])
24 avg
25
26 #checks null value in data
27 print(combine.isnull().sum())
28

```

PROBLEMS

OUTPUT

TERMINAL

...

Python + ▾ □ 🗑️ ...

```

0      1  john      10      85
1      2  sara      20      90
3      4  anna      30      95
>>> avg = stud.groupby('dept_id')['marks'].agg(['mean'])
>>> avg
      mean
dept_id
10      77.5
20      90.0
30      95.0
40      60.0
>>> 

```

```
25
26 #checks null value in data
27 print(combine.isnull().sum())
28
29 #checks duplicate values in data
30 print(combine.duplicated().sum())
31
32 #checks stats
33 print(combine.describe())
```

PROBLEMS

OUTPUT

TERMINAL

...

Python + -



```
>>> print(combine.isnull().sum())
stud_1d      0
name         0
dept_id      0
marks        0
dept_name    0
dtype: int64
>>> #checks duplicate values in data
>>>
>>> print(combine.duplicated().sum())
0
>>> #checks stats
>>>
>>> print(combine.describe())
```

	stud_1d	dept_id	marks
count	4.000000	4.000000	4.000000
mean	2.500000	17.500000	85.000000
std	1.290994	9.574271	10.801234
min	1.000000	10.000000	70.000000
25%	1.750000	10.000000	81.250000
50%	2.500000	15.000000	87.500000